

torch.linalg.matrix_exp#

https://pytorch.org/docs/stable/generated/torch.linalg.matrix_exp.html

Description:

Computes the matrix exponential of a square matrix. Letting $K \in \mathbb{R} \text{ or } \mathbb{C}$, this function computes the matrix exponential of $A \in K^{n \times n}$, which is defined as If the matrix A has eigenvalues $\lambda_i \in \mathbb{C}$, the matrix $\text{matrix_exp}(A)$ has eigenvalues $e^{\lambda_i} \in \mathbb{C}$. Supports input of `bfloat16`, `float`, `double`, `cfloat` and `cdouble` dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions. A (Tensor) – tensor of shape $(*, n, n)$ where $*$ is zero or more batch dimensions.

Function Signature

`torch.linalg.matrix_exp(A) → Tensor#`

Parameters

Code Examples

```
>>> A = torch.empty(2, 2, 2)
>>> A[0, :, :] = torch.eye(2, 2)
>>> A[1, :, :] = 2 * torch.eye(2, 2)
>>> A
tensor([[[1., 0.],
         [0., 1.]],

        [[2., 0.],
         [0., 2.]]])
>>> torch.linalg.matrix_exp(A)
tensor([[[2.7183, 0.0000],
         [0.0000, 2.7183]],

        [[7.3891, 0.0000],
         [0.0000, 7.3891]]])

>>> import math
>>> A = torch.tensor([[0, math.pi/3], [-math.pi/3, 0]]) # A is
skew-symmetric
>>> torch.linalg.matrix_exp(A) # matrix_exp(A) = [[cos(pi/3),
sin(pi/3)], [-sin(pi/3), cos(pi/3)]]
tensor([[ 0.5000,  0.8660],
        [-0.8660,  0.5000]])
```

Detailed Documentation

Documentation

Computes the matrix exponential of a square matrix.

Letting K be $\mathbb{R}$ or $\mathbb{C}$, this function computes the matrix exponential of $A \in K^{n \times n}$, which is defined as

If the matrix A has eigenvalues $\lambda_i \in \mathbb{C}$, the matrix $\text{matrix_exp}(A)$ has eigenvalues $e^{\lambda_i} \in \mathbb{C}$.

Supports input of `bfloat16`, `float`, `double`, `cfloat` and `cdouble` dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

A (Tensor) – tensor of shape $(*, n, n)$ where $*$ is zero or more batch dimensions.